ER Diagrams

Question 1:

A university registrar's office maintains data about the following entities:

1. courses, including number, title, credits, syllabus, and prerequisites;
2. course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom;
3. students, including student-id, name, and program;
4. Instructors, including identification number, name, department, and title.

Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the mapping constraints.

Question 2:

Consider a database used to record the marks that students get in different exams of different course offerings. Construct the ER diagram.

Question 3:

Draw the E-R diagram which models an online bookstore.

- Books are written by at least 1 author and published by only 1 publisher. Books have year, title, price, ISBN.
- Customers have a shopping cart, where they can add books. They can add more than one book of the same title at the same time.
- The books are stored in a warehouse; there are multiple warehouses in different locations.

Question 4:

Consider a university database for the scheduling of classrooms for -final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as

- course with attributes name, department, and c-number
- section with attributes s-number and enrollment, and dependent as a weak entity set on course
- room with attributes r-number, capacity, and building

Show an E-R diagram illustrating the use of all three additional entity sets listed.